

Certificate of Analysis

PRODUCT DETAILS

Name

Batch

Date

Cell Source and Passage

Human Umbilical Cord (Wharton's Jelly), Passage 2

Storage Temperature

4 – 8°C

Viable Cell Count

TEST	SPECIFICATION	TEST RESULTS	REMARKS
Infectious Diseases Screening	HBsAg; HCV Ab; HIV1/2 Ag/Ab; CMV Ab; RPR	Not Detected / Non-Reactive	QUALITY CONTROL APPROVED
Morphology of Cells	Spindle-like cells	Small spindle shaped cells	QUALITY CONTROL APPROVED
Cell Viability	$\geq 90\% \pm 0.1$ of requested cells	Within range	QUALITY CONTROL APPROVED
Sterility	No cloudiness of the test media	Negative	QUALITY CONTROL APPROVED
Bacterial Endotoxin Test (LAL)	Endotoxin sensitivity 0.125 EU/mL	Negative	QUALITY CONTROL APPROVED
Mycoplasma	Mean of absorbance <0.05	Negative	QUALITY CONTROL APPROVED
Immunophenotyping	Positive marker – CD73, CD90, CD105 ($\geq 95\%$) Negative marker – CD11b, CD19, CD34, CD45, HLA-DR ($\leq 2\%$)	Within range	QUALITY CONTROL APPROVED
Differentiation Capacity	Minimum tri-lineage differentiation	Positive stain on differentiated cells	QUALITY CONTROL APPROVED
Karyotyping	Normal cells = 46 chromosomes	Normal	QUALITY CONTROL APPROVED

The cell product delivered by Malaysian Genomics Resource Centre Berhad is manufactured and tested for research purpose only. Product is processed in a facility operating under current Good Manufacturing Practice (cGMP) guidelines in accordance with PIC/S standards.

Authorised Signature:

Quality Control Department

Certificate of Analysis

Supporting Information

MORPHOLOGY OF CELLS

Figure 1: Mesenchymal stem cells as viewed under an inverted phase contrast microscope (10x).

STERILITY TEST ^{6, 8}

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.

Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

BACTERIAL ENDOTOXIN TEST (LIMULUS AMOEBOCYTE LYSATE, LAL) TEST ^{3,6}

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

MYCOPLASMA TEST ^{4,6}

Negative findings for mycoplasma species *M. hyorhinis*, *M. arginini*, *M. fermentans*, *M. orale*, *M. Pirum*, *M. hominis*, *M. salivarium* and *A. laidlawii*.

IMMUNOPHENOTYPING

Marker	Positive			Negative (CD11b, CD19, CD34, CD45, HLA-DR)
	CD73+	CD90+	CD105+	
Result (%)				

Table 1: Flow cytometric analysis was performed to verify the characteristic surface marker expression profile of mesenchymal stem cells.

DIFFERENTIATION CAPACITY ¹

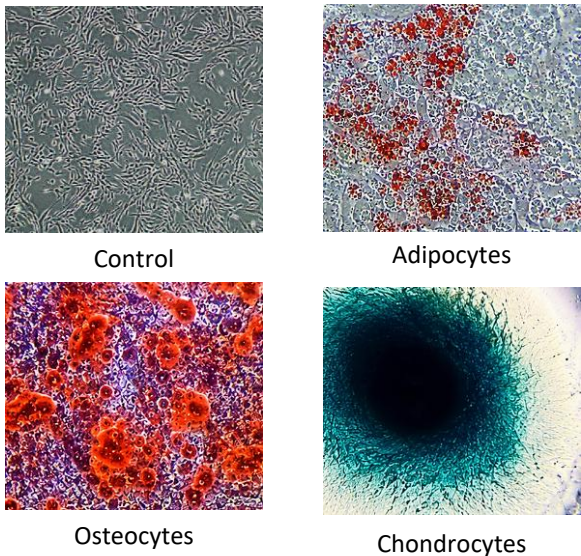


Figure 2: In-vitro differentiated stem cells observed under an inverted phase contrast microscope (10x). The cells demonstrated successful differentiation into adipocytes, osteocytes and chondrocytes.

KARYOTYPING ⁷

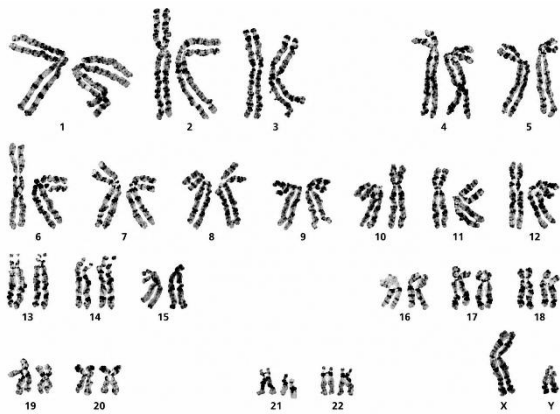


Figure 3: Karyotype analysis of the mesenchymal stem cells demonstrated a normal chromosomal complement of 46, XY. No numerical or structural chromosomal abnormalities were detected. The cells exhibit a normal diploid karyotype.



REFERENCES

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5. USP Method: Sterilization of Medical Devices, Microbiological Methods.
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